ENGINEERING ECONOMICS HOMEWORK

IRR (i*), Present Worth, and Incremental Rate of Return

Given. The minimum attractive rate of return is

$$\text{MARR} = 15\% \text{ per year.}$$

Two mutually exclusive alternatives, **A** and **B**, are available. You are assigned *one* of the data sets below (Set 1–Set 4). All cash flows are in dollars ($) and occur at the end of each year unless otherwise stated.

Required. For your assigned data set, answer the following questions using Excel.

1. Compute the internal rate of return, IRR_A, for alternative A.
2. Compute the internal rate of return, IRR_B, for alternative B.
3. Based on IRR values alone, determine whether each alternative is individually acceptable at the given MARR.
4. Compute the present worth PW_A of alternative A at MARR.
5. Compute the present worth PW_B of alternative B at MARR.
6. Using the present worth method, determine which alternative should be selected.
7. Construct the incremental cash flow series $(B - A)$.
8. Compute the incremental present worth PW_{B-A} at MARR.
9. Interpret the sign of PW_{B-A} and state whether the additional investment in B over A is economically justified.
10. Compute the incremental internal rate of return, IRR_{B-A}.
11. Compare IRR_{B-A} with MARR and make the final decision.
12. Briefly explain your decision using incremental analysis.

Data Sets

Set 1

Year	Alternative A	Alternative B
0	-10,000	-15,000
1	3,500	4,500
2	3,500	4,500
3	3,500	4,500
4	3,500	4,500
5	3,500	8,000

Set 2

Year	Alternative A	Alternative B
0	-12,000	-17,000
1	4,200	4,700
2	4,200	4,700
3	4,200	4,700
4	4,200	4,700
5	4,200	7,700

Set 3

Year	Alternative A	Alternative B
0	-9,000	-14,000
1	3,000	4,200
2	3,000	4,200
3	3,000	4,200
4	3,000	4,200
5	3,000	7,200

Set 4

Year	Alternative A	Alternative B
0	-9,000	-15,000
1	3,500	4,800
2	3,500	4,800
3	3,500	4,800
4	3,500	4,800
5	3,500	5,800

Note. Clearly state which data set you are solving (Set 1, 2, 3, or 4). All calculations must be performed using Excel.